Topics: Reflection, Protected Internal, and Private Protected

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1. Reflection in C#

Definition

Reflection is a feature in C# that allows a program to examine information about its own types, classes, methods, properties, and objects at runtime.

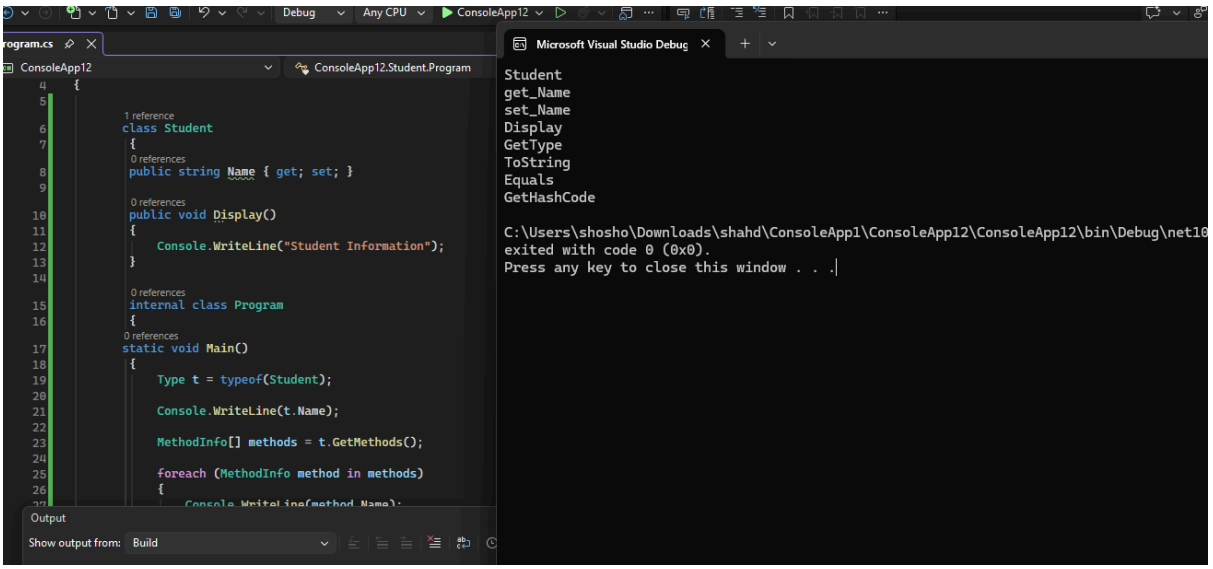
Explanation

Reflection is provided by the `System.Reflection` namespace. It allows developers to discover information about classes and their members while the program is running.

It can be used to:

- Get information about classes and objects.
- Examine methods, properties, and fields.
- Create objects dynamically.
- Call methods at runtime.
- Read metadata about types.

Example



2. Protected Internal in C#

Definition

`protected internal` is an access modifier in C# that allows a class member to be accessed from the same assembly or from derived classes in other assemblies.

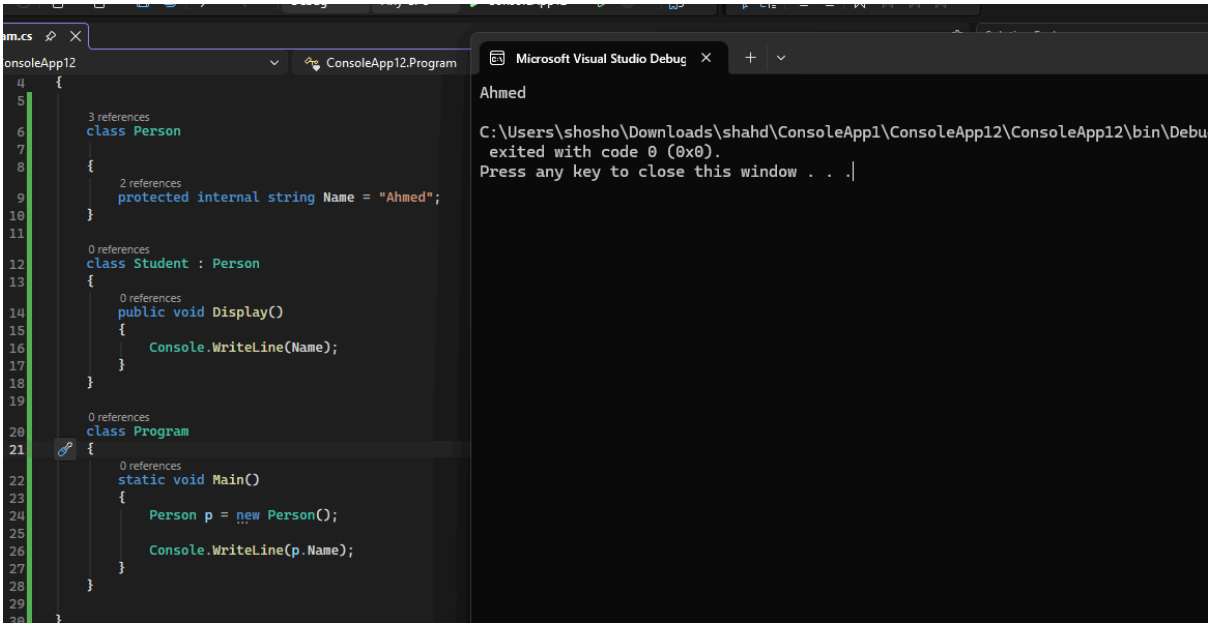
Explanation

The `protected internal` access modifier combines two access levels:

- **Protected:** Accessible within the same class and its derived classes.
- **Internal:** Accessible anywhere within the same assembly.

This means that the member can be accessed if either condition is satisfied.

Example



3. Private Protected in C#

Definition

`private protected` is an access modifier in C# that allows a class member to be accessed only within the same assembly and by derived classes.

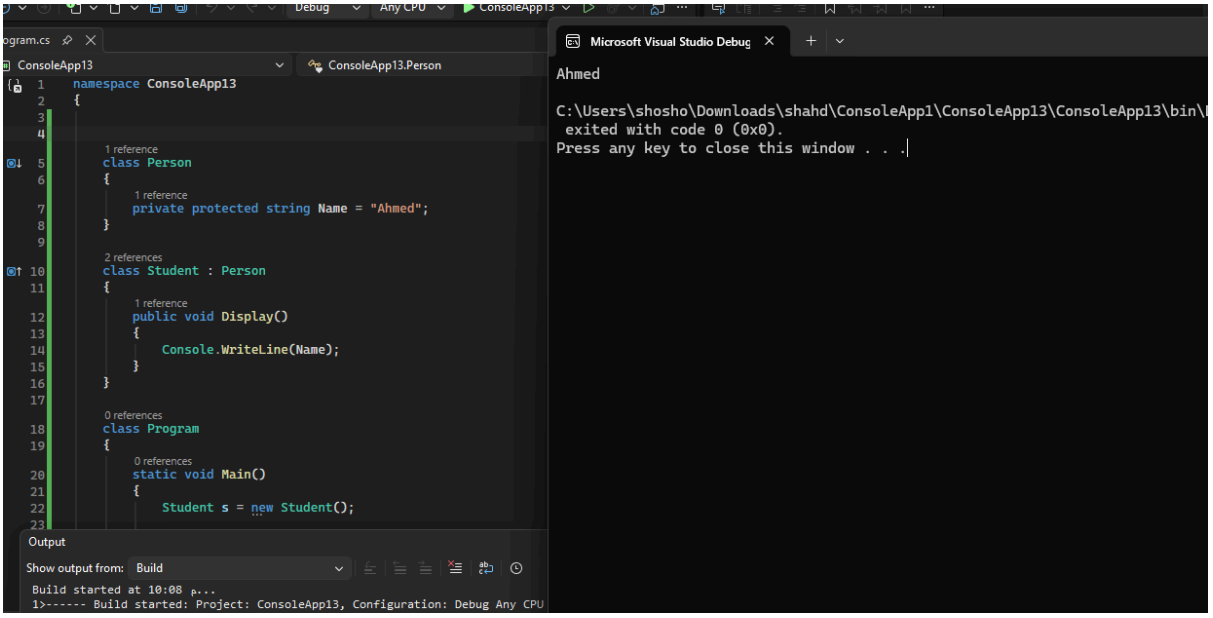
Explanation

The `private protected` access modifier combines two access restrictions:

- **Private:** Restricts access to the containing class.
- **Protected:** Allows access to derived classes.

A `private protected` member can be accessed by derived classes only when they are in the same assembly as the containing class.

Example



Comparison Between Protected Internal and Private Protected

Feature	Protected Internal	Private Protected
Same Assembly	Yes	Yes, through a derived class

Derived Class in Same Assembly	Yes	Yes
Derived Class in Another Assembly	Yes	No
Access Level	More flexible	More restricted